



Filter Wizard

Filter Wizard Design

Created on 06/22/2025



Filter Wizard Design Report

Filter Requirements for Low-Pass, 4th order Butterworth

Specifications: Optimize: Specific Parts; +Vs: 5; -Vs: -5

Gain: 0 dB

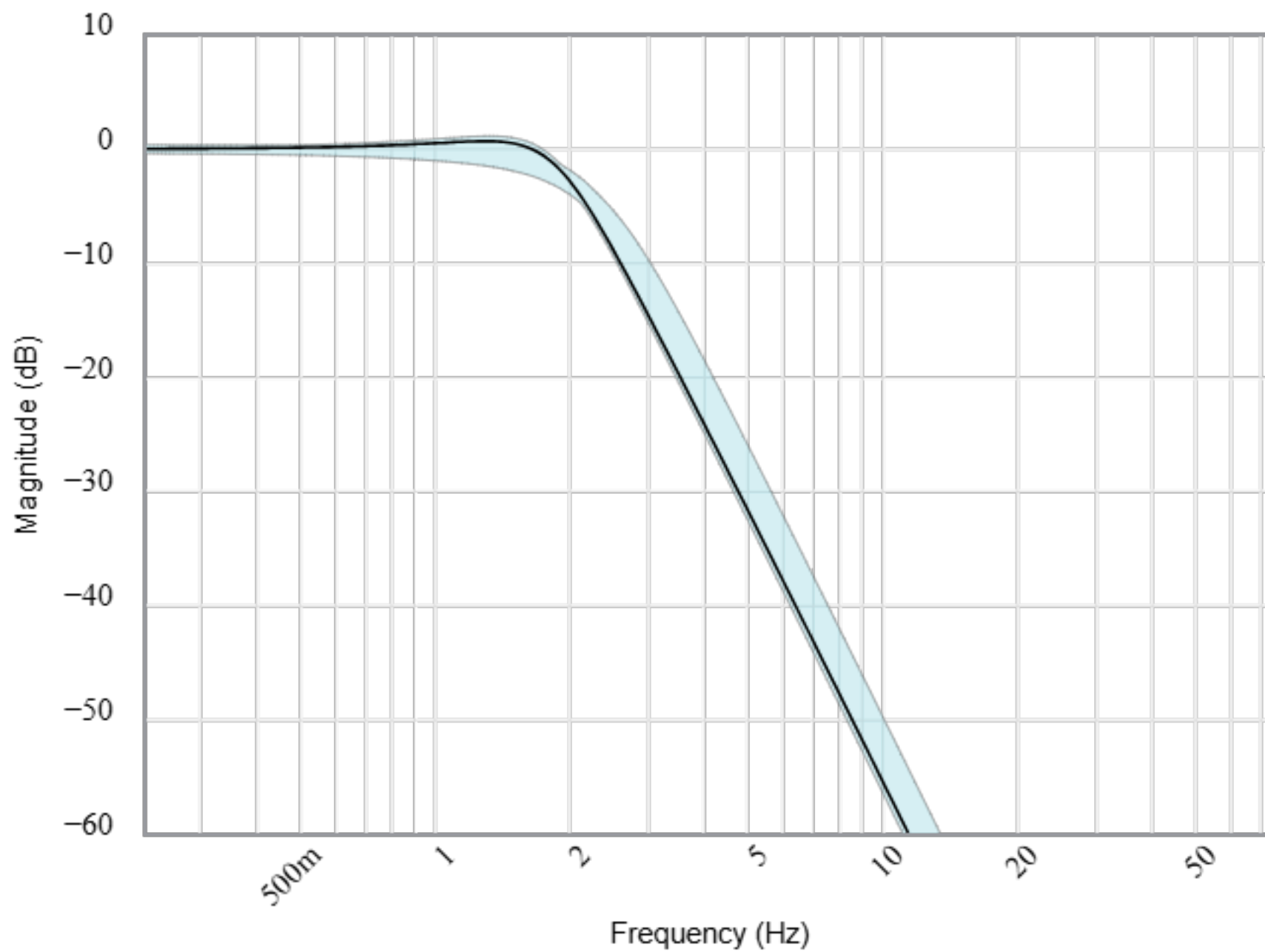
Passband: -3dB at 2.3Hz

Stopband: -40dB at 7.5Hz

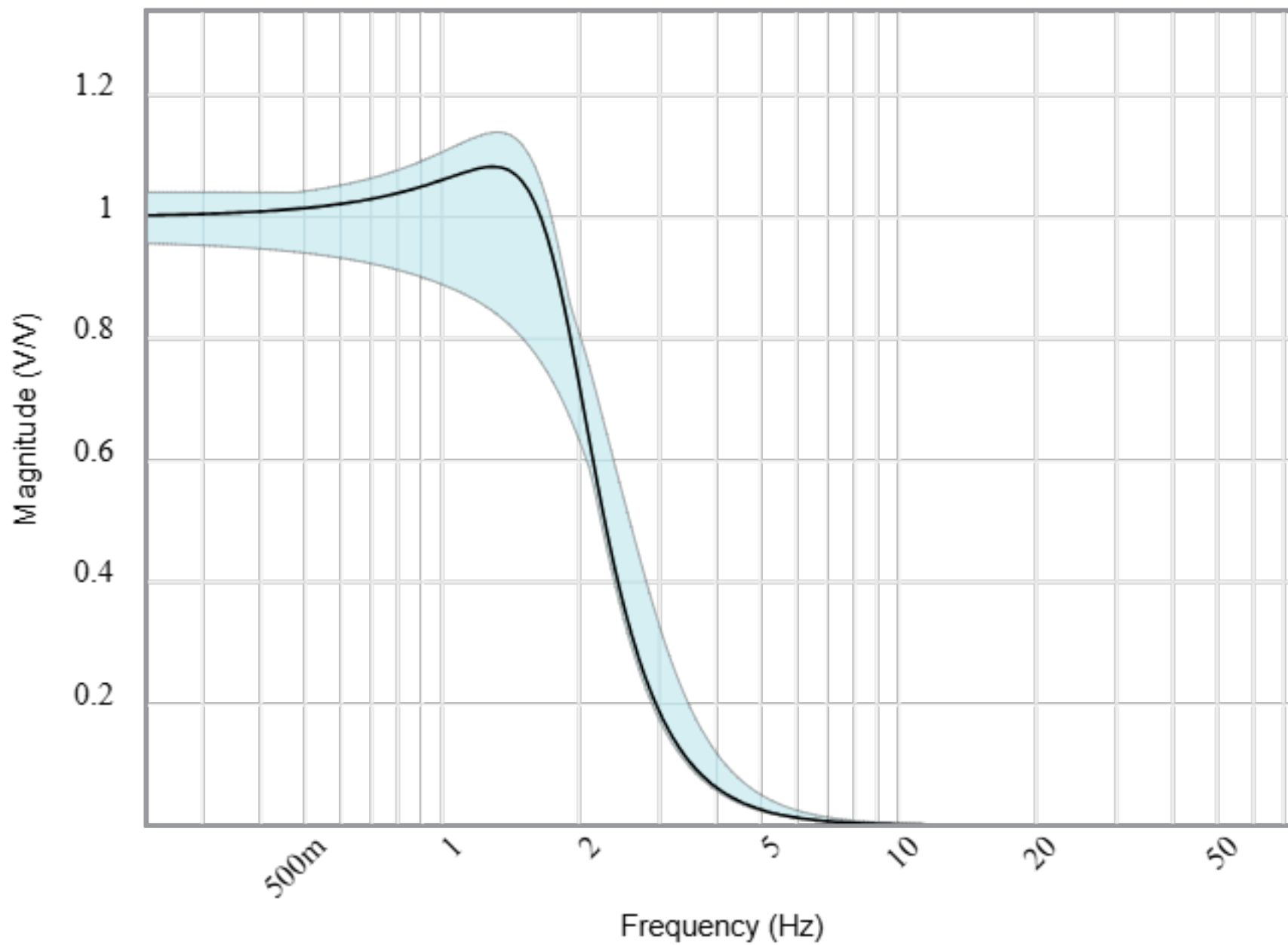
Component Tolerances: Capacitor = 5%; Resistor = 1%; Inductor = 5%; Op Amp GBW = 20%

BOM: refer to BOM.csv file

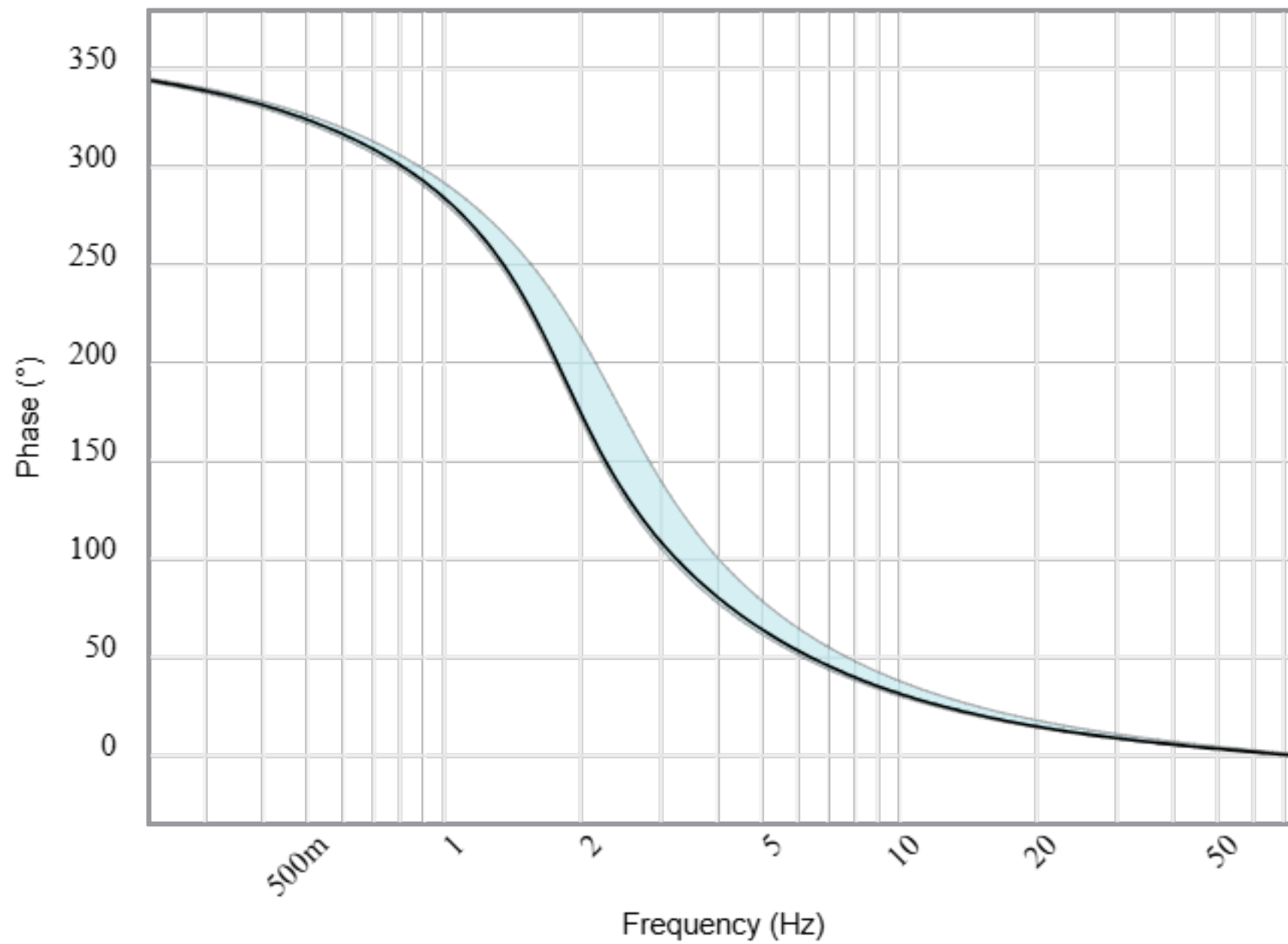
Magnitude(dB)



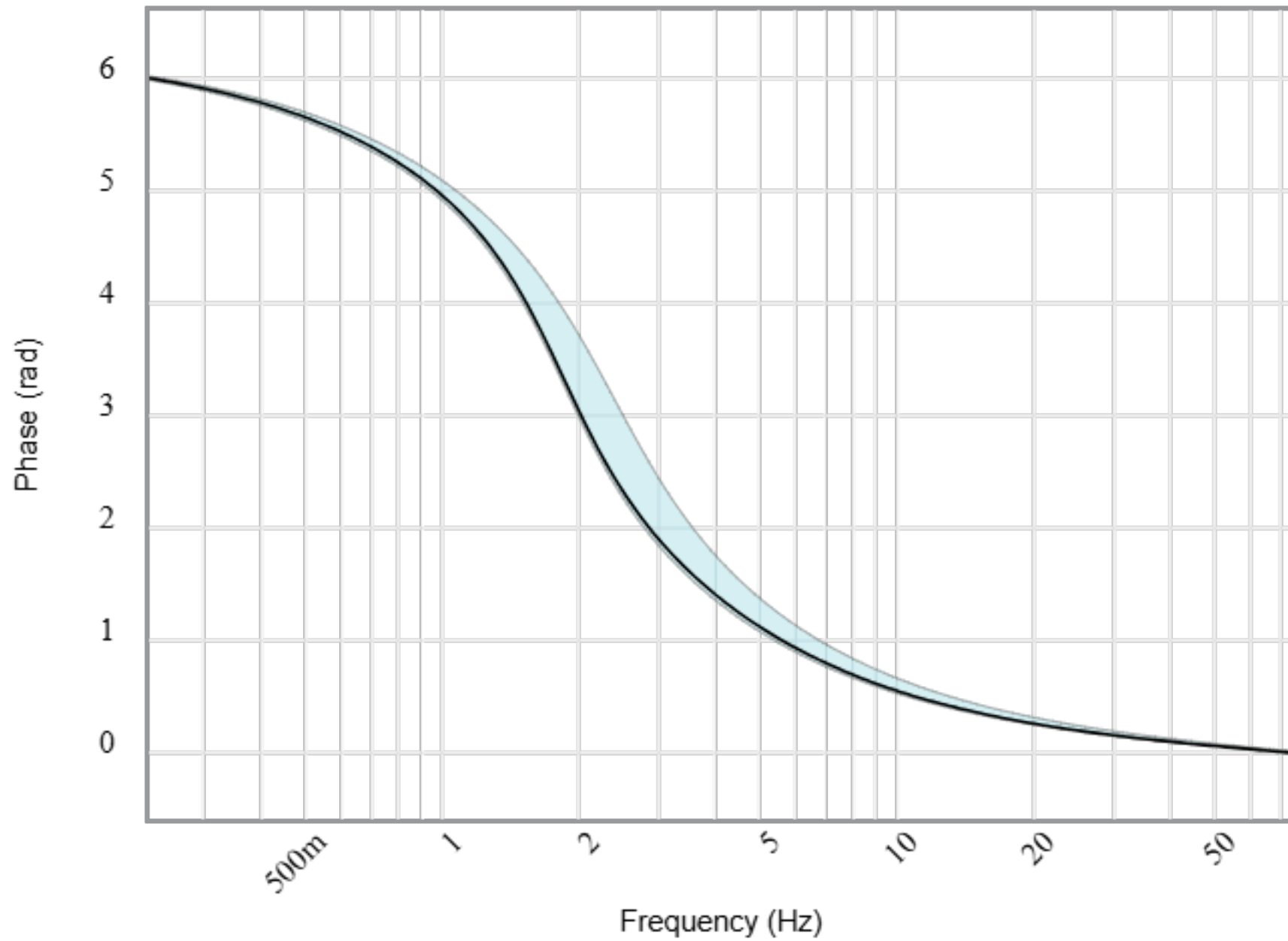
Magnitude(Volts per Volt)



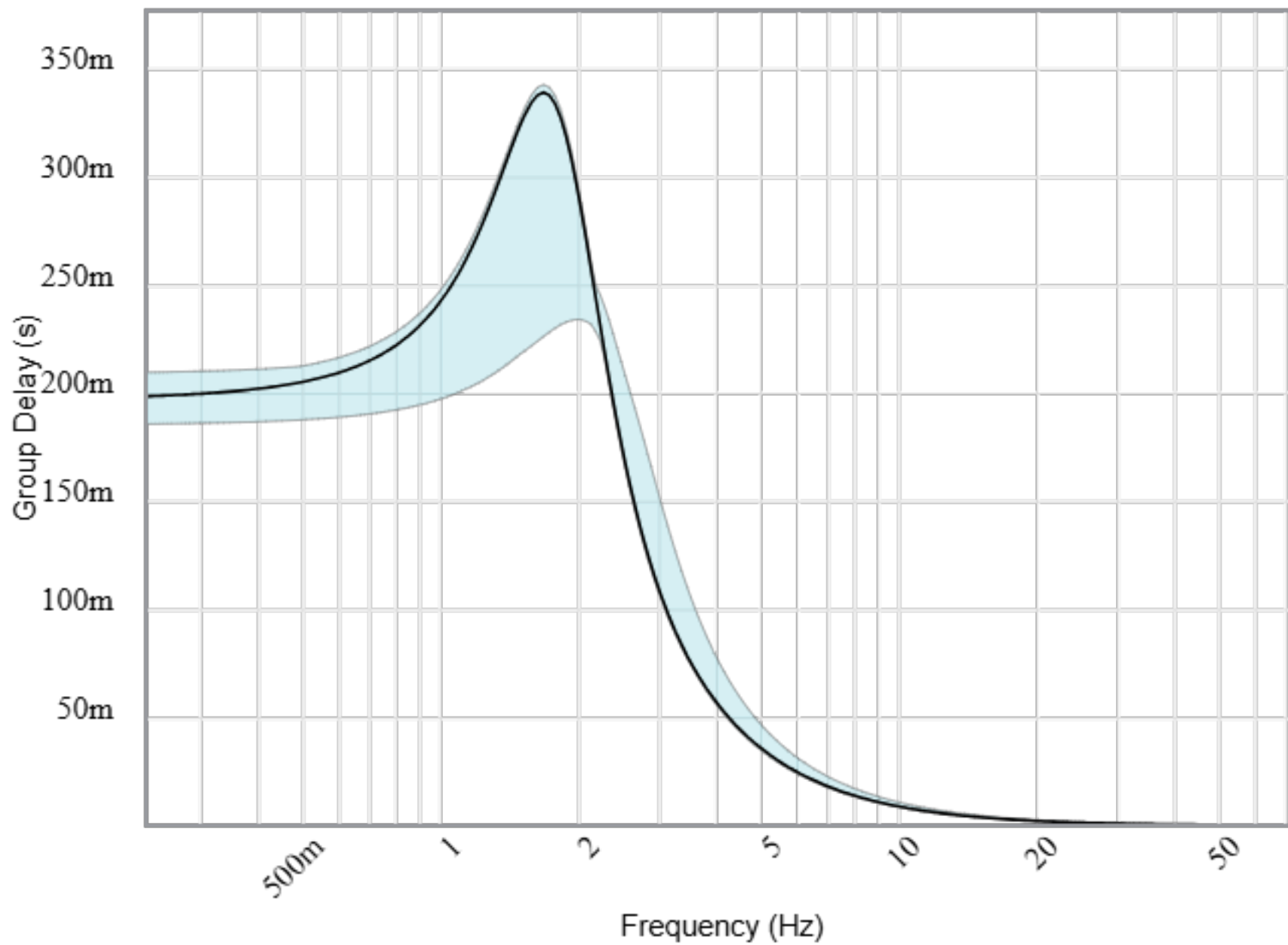
Phase(degrees)



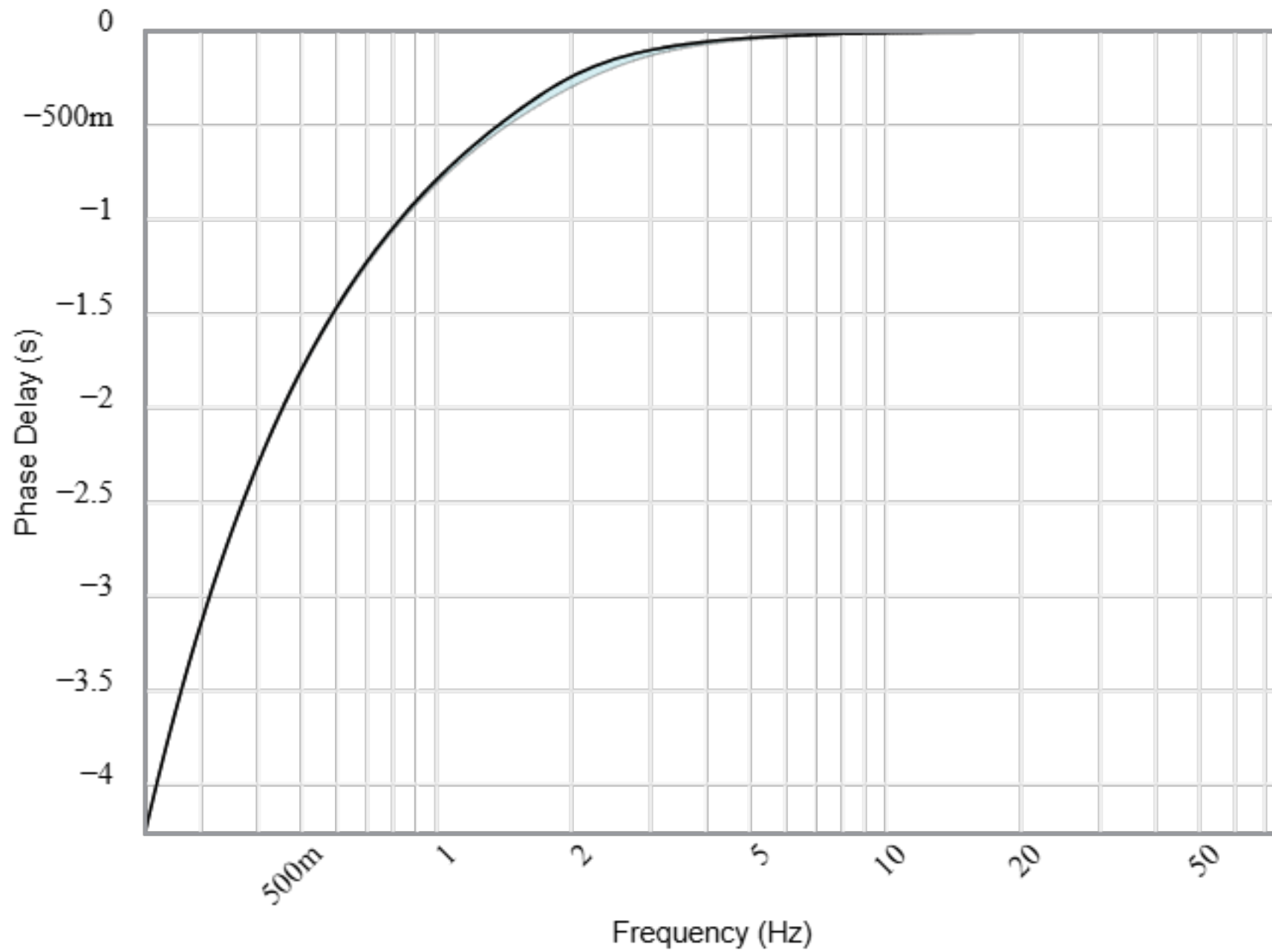
Phase(radians)



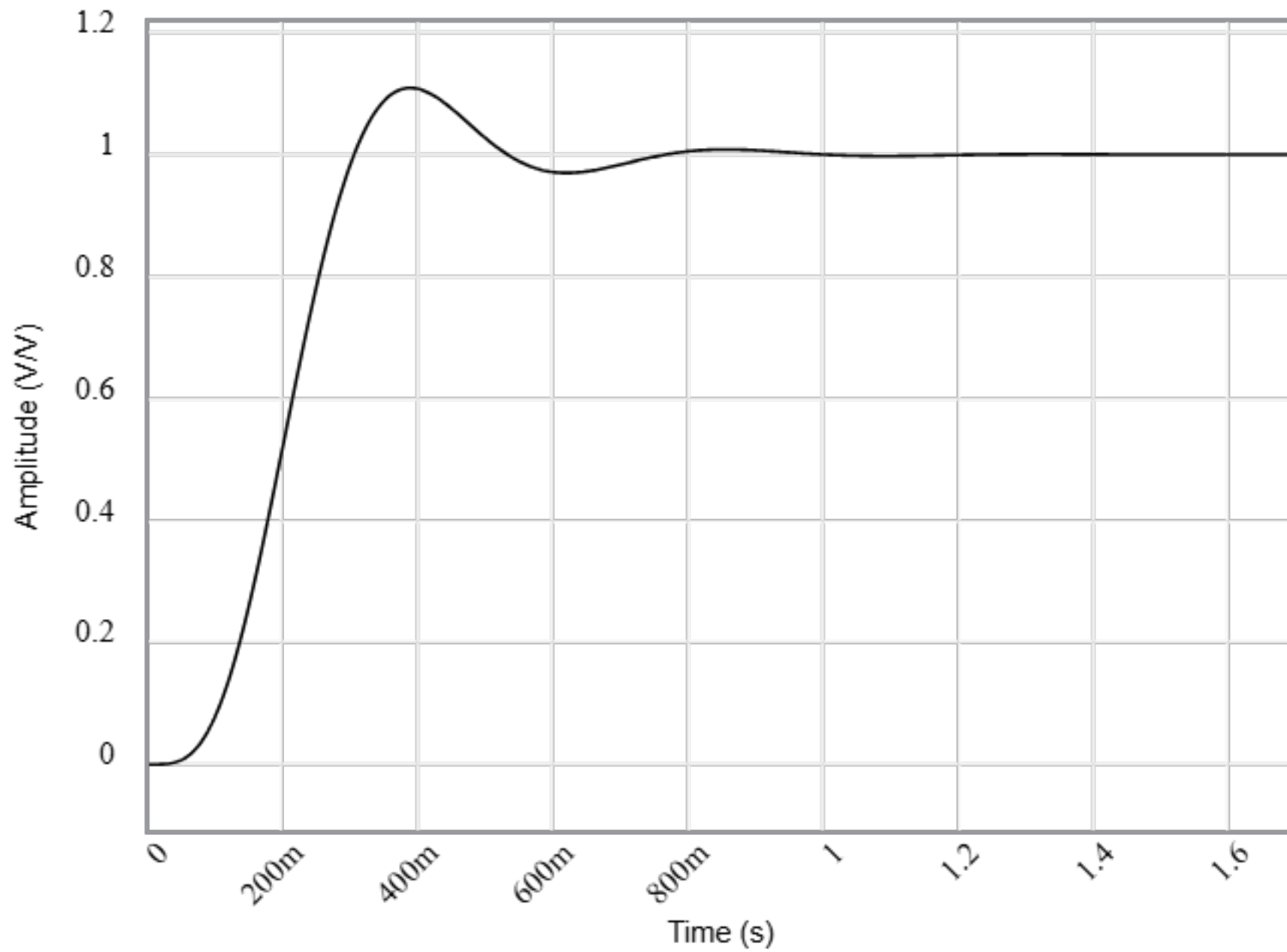
Group Delay



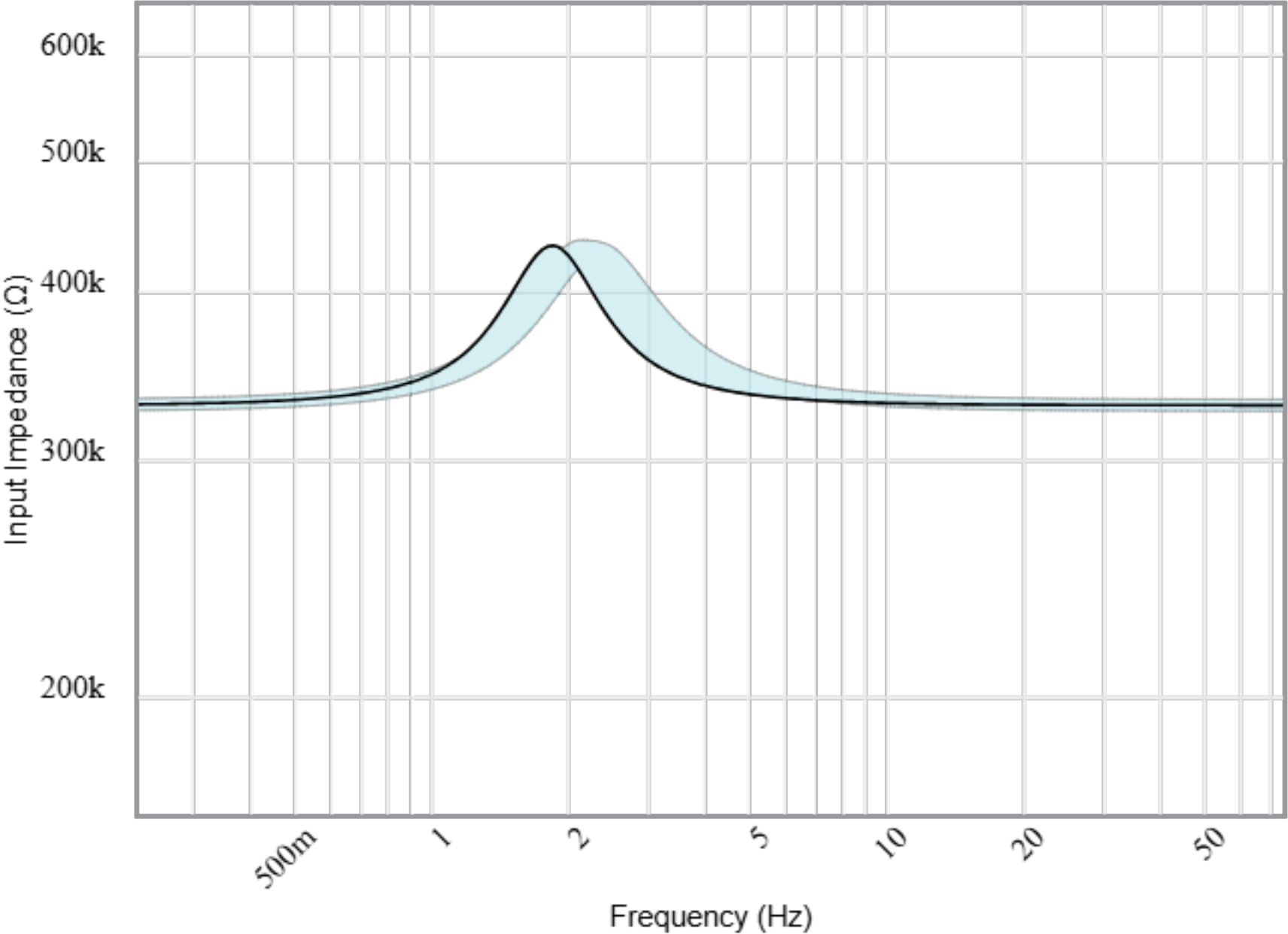
Phase Delay



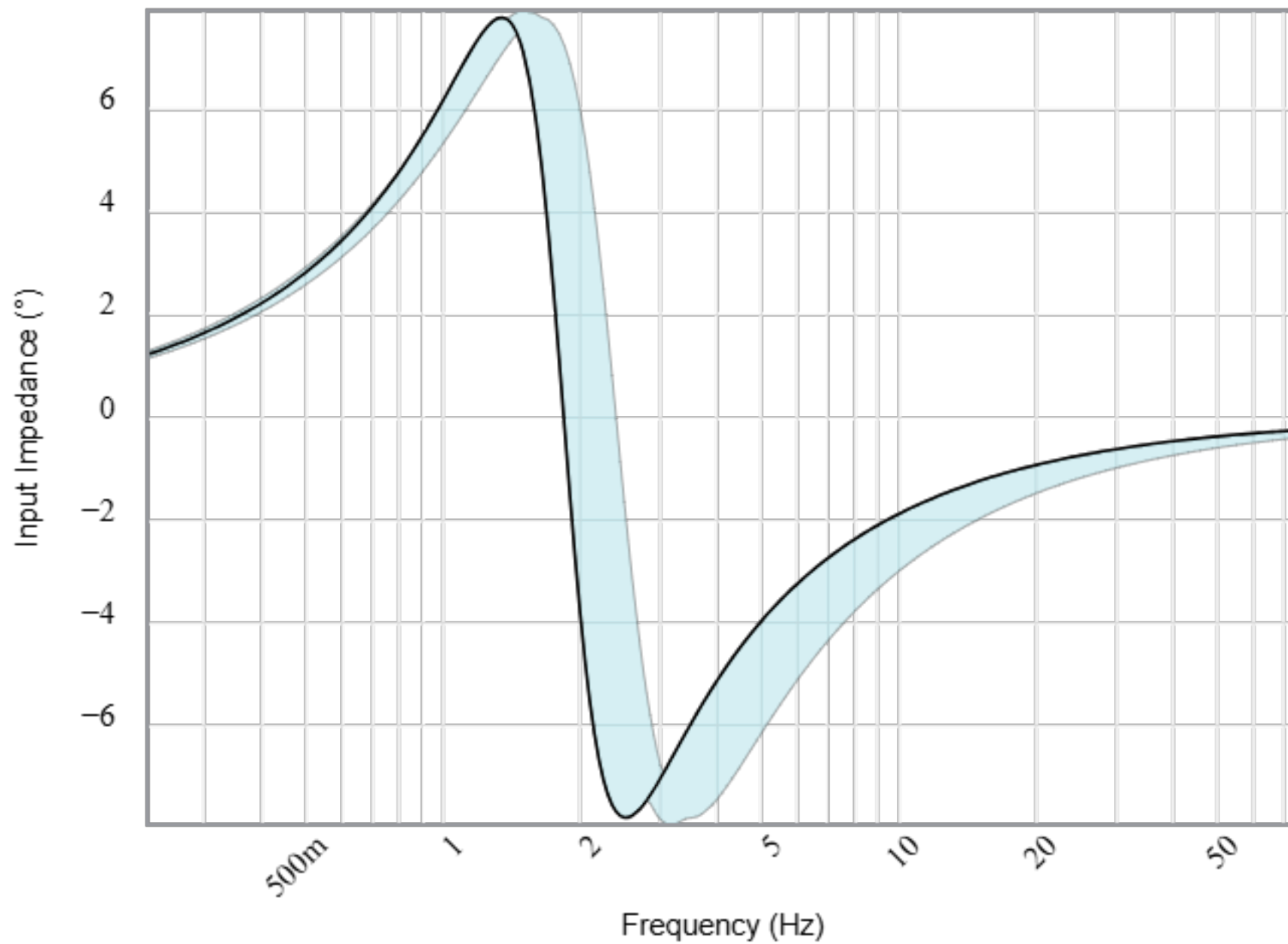
Step Response



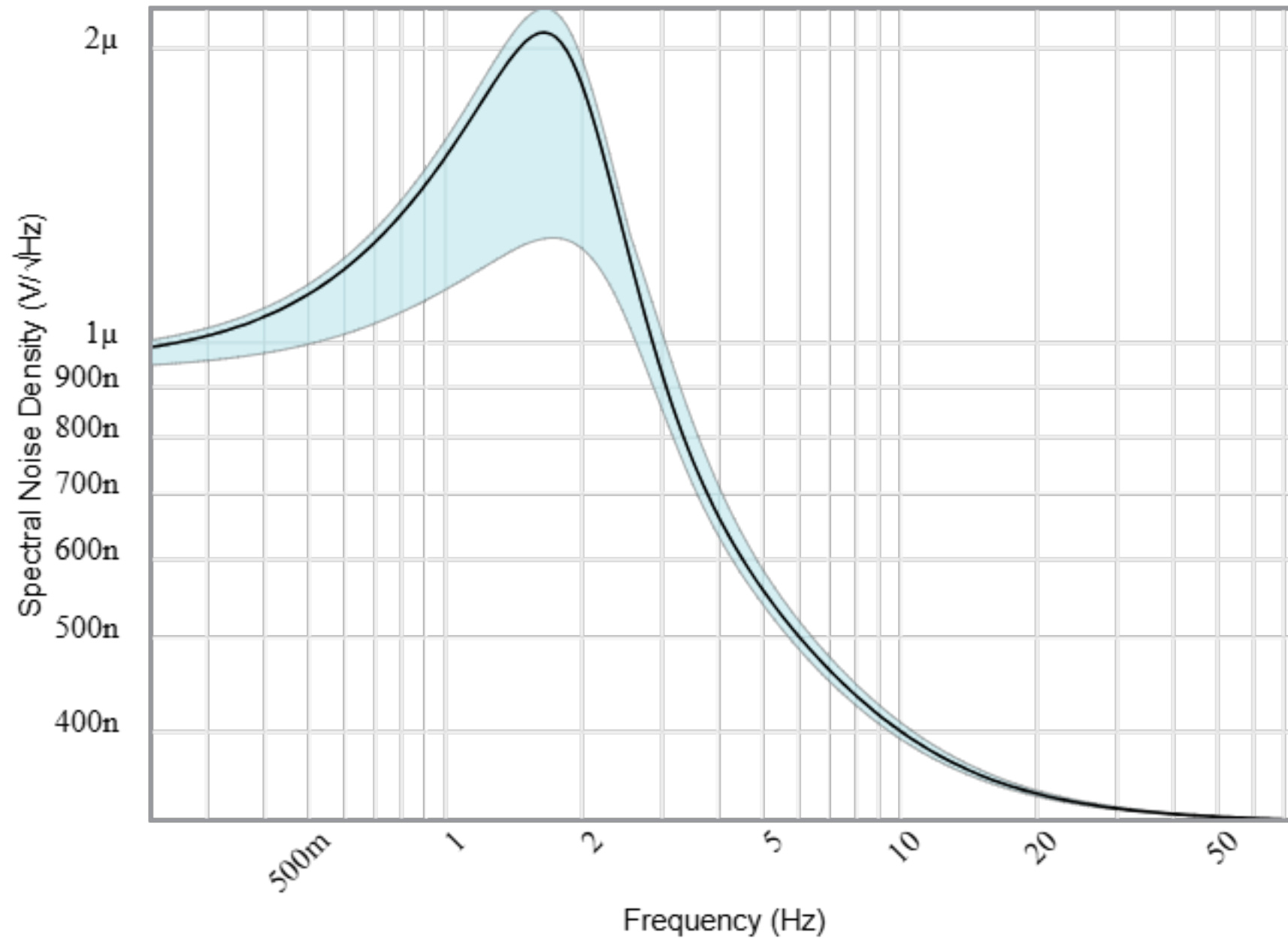
Input Impedance Magnitude



Input Impedance Phase



Noise



Stages

Your filter requires 2 op amp stage(s) with the following characteristics



**2nd order
Low-Pass
Multiple
Feedback**

Target

Simulated

Gain (V/V):

1

0.98 to 1.02

f_p (Hz):

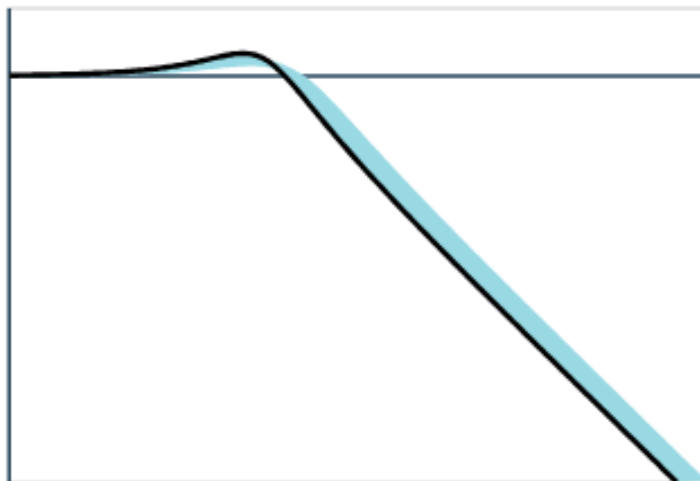
2.3

2.13 to 2.4

Q:

1.31

1.07 to 1.31



**2nd order
Low-Pass
Multiple
Feedback**

Target

Simulated

1

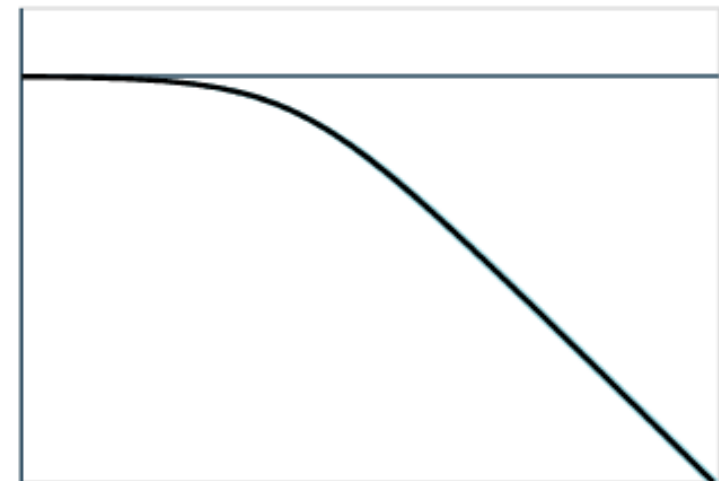
0.98 to 1.02

2.3

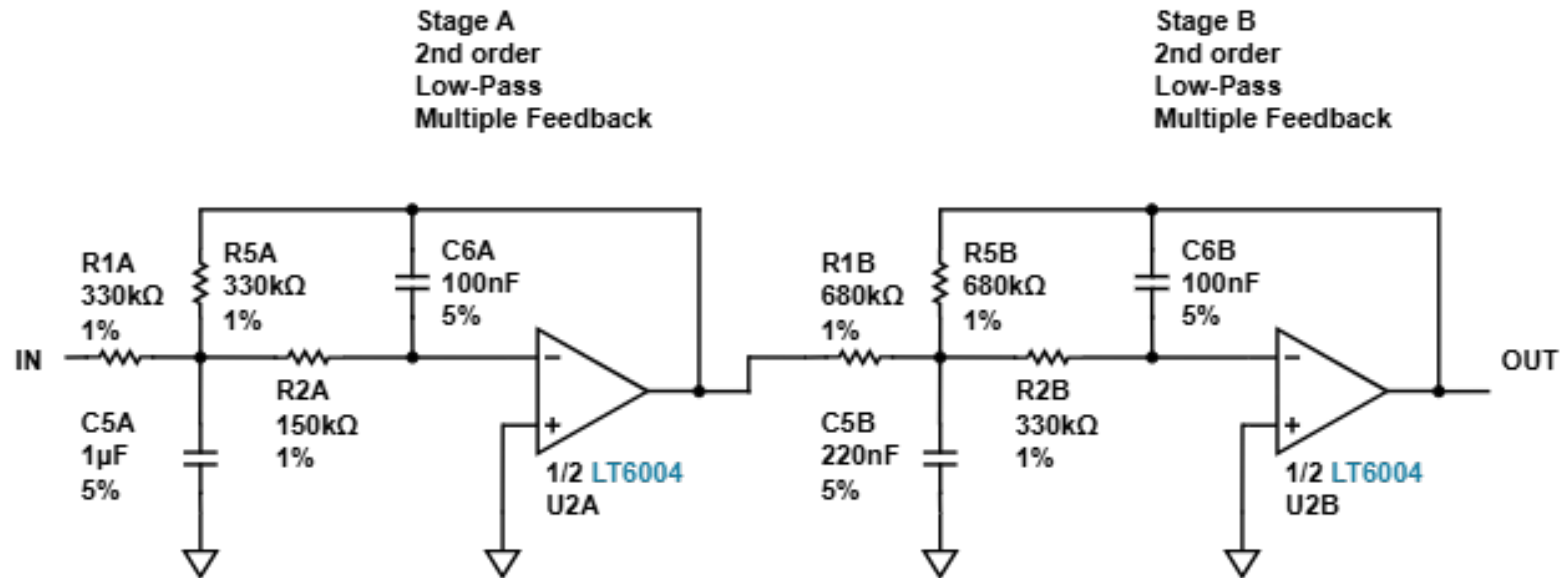
2.13 to 2.4

541m

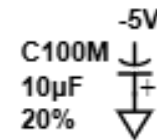
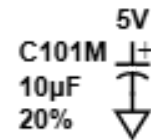
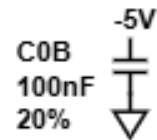
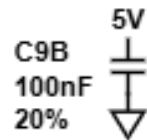
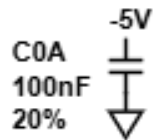
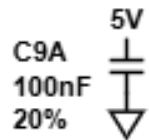
501m to 549m



Circuit



BYPASS CAPACITORS



SPARES Why The Spares?

